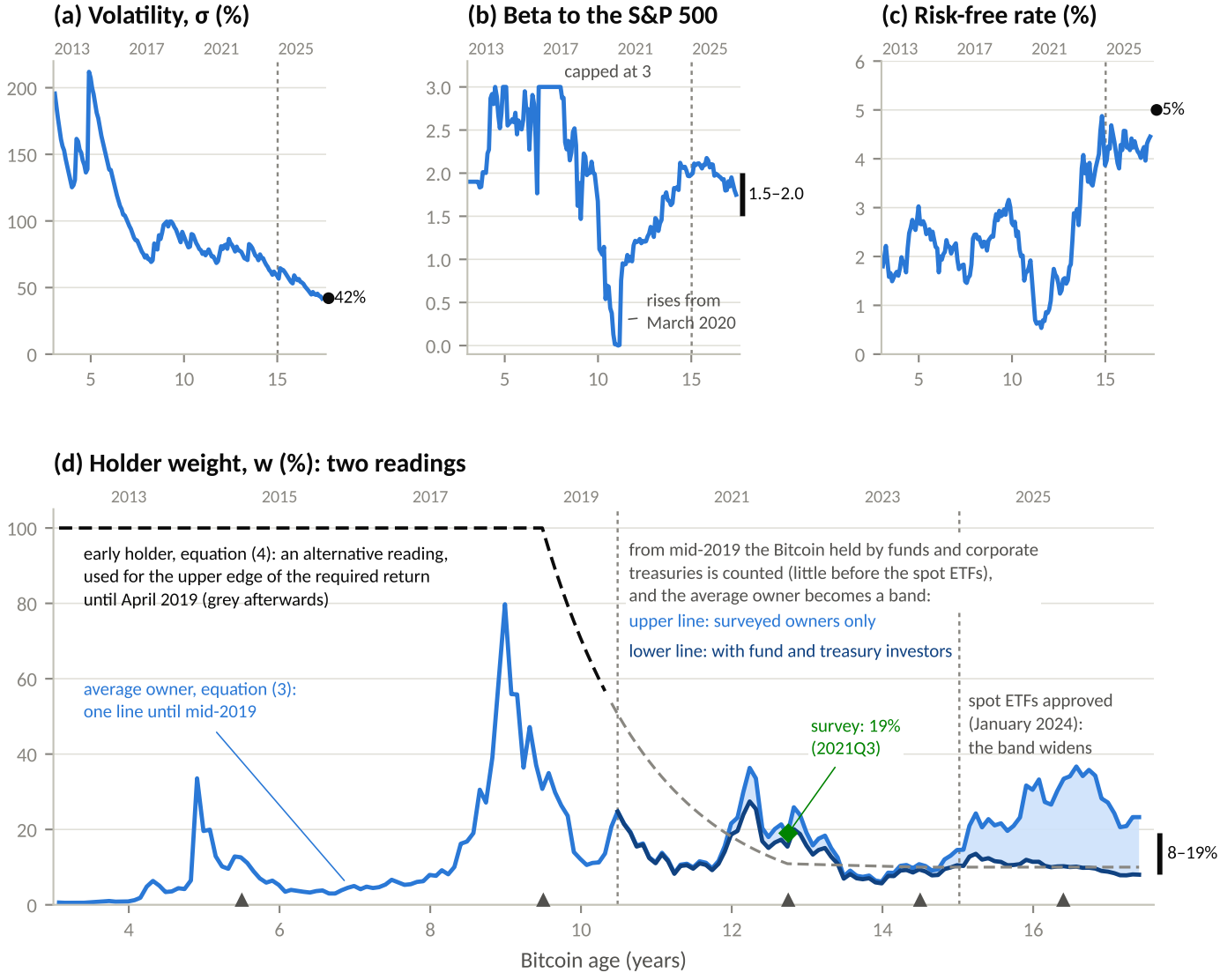

(a) Volatility, σ (%)
2013
2017
2021
2025
200
150
100
50
0
5
10
15
42%
(b) Beta to the S&P 500
capped at 3
3.0
2.5
2.0
1.5
1.0
0.5
0.0
1.5–2.0
rises from
March 2020
(c) Risk-free rate (%)
6
5
4
3
2
1
0
5%
(d) Holder weight, w (%): two readings
2013
2015
2017
2019
2021
2023
2025
100
80
60
40
20
0
early holder, equation (4): an alternative reading,
used for the upper edge of the required return
until April 2019 (grey afterwards)
from mid-2019 the Bitcoin held by funds and corporate
treasuries is counted (little before the spot ETFs),
and the average owner becomes a band:
upper line: surveyed owners only
lower line: with fund and treasury investors
average owner, equation (3):
one line until mid-2019
survey: 19%
(2021Q3)
spot ETFs approved
(January 2024):
the band widens
8–19%
4
6
8
10
12
14
16
Bitcoin age (years)